Session 3 — Working Capital Management

By the end of this session you must be able to compute and interpret DSO, DPO and DI, derive the cash conversion cycle from them, explain the carrying-cost vs order-cost trade-off in inventory management, and walk through the four questions that structure receivables management (terms of sale, credit analysis, the credit decision, collection policy).

Plan

1. The Cash Conversion Cycle
2. Managing Inventories: The Carrying-Cost vs Order-Cost Trade-off
3. Receivables Management I: Terms of Sale and Credit Analysis
4. Receivables Management II: The Credit Decision and Collection Policy
5. Common Mistakes & Exam Traps
6. Must Know Absolutely

1. The Cash Conversion Cycle

1.1 Definitions: Inventory, Receivable and Payable Periods

> **The cash cycle:** the firm buys raw materials, processes them into finished goods, and generally sells them on credit. The cash cycle measures the net delay between paying cash out for inputs and collecting cash in from customers, once the delay the firm itself gets from its own suppliers is netted out.

Three periods make up the cycle: the **inventory period** (delay between the initial investment in inventory and the final sale); the **receivable period** (delay between the sale of goods and the customer's payment); and the **accounts payable period** (delay between the purchase of raw materials and payment for them). The inventory period plus the receivable period together form the **operating cycle** — the full stretch from purchasing raw materials to collecting cash on the sale. The **cash cycle** is shorter than the operating cycle by however long the firm's own suppliers finance it: cash cycle = operating cycle − accounts payable period.

$$\text{DSO} = \text{Accounts Receivable} \div (\text{Sales} \div 365)$$
$$\text{DPO} = \text{Accounts Payable} \div (\text{COGS} \div 365)$$
$$\text{DI} = \text{Inventory} \div (\text{COGS} \div 365)$$
$$\text{Cash Cycle} = \text{DI} + \text{DSO} - \text{DPO}$$

where:

- DSO (days sales outstanding) is the average receivable period; DPO (days payables outstanding) is the average payable period; DI (days inventory) is the inventory period.
- DSO is scaled by daily **sales**; DI and DPO are both scaled by daily **COGS**, not sales — a frequent source of calculation errors.
- Cash Cycle = DI + DSO − DPO = operating cycle (DI + DSO) minus the payable period, since payables effectively finance part of the operating cycle.

1.2 Worked Example: Computing DSO, DPO, DI and the Cash Cycle

Example (solved in source, $ mil, FY2025): Sales $6,133. COGS $5,464. Inventory $879. Accounts Receivable $764. Accounts Payable $682.
DSO = 764 ÷ (6,133 ÷ 365) = 45.468775 ≈ 45 days.
DPO = 682 ÷ (5,464 ÷ 365) = 45.558199 ≈ 46 days.
DI = 879 ÷ (5,464 ÷ 365) = 58.717972 ≈ 59 days.
Applying the formula to the unrounded values: Cash Cycle = 58.717972 + 45.468775 − 45.558199 = 58.628549 ≈ 59 days, the answer the slide shows. Keep the full decimals until the final result. Summing the periods already rounded (59 + 45 − 46 = 58) is only correct if an exam statement hands you the three periods pre-rounded as its data.

Worked example (self-test data, applying the same formulas): Walmart — Sales $514, COGS $385, Inventory $44, AR $6, AP $46 → DSO ≈ 4 days, DI ≈ 42 days, DPO ≈ 44 days, Cash Cycle ≈ 2 days. Boeing — Sales $101, COGS $82, Inventory $44, AR $10, AP $12 → DSO ≈ 36 days, DI ≈ 196 days, DPO ≈ 53 days, Cash Cycle ≈ 179 days. The contrast illustrates the formula well: a fast-turnover retailer (Walmart) has a cash cycle near zero, while a long-manufacturing-cycle industrial firm (Boeing) has one of nearly six months.

Watch out: the source's own solved example states the cash cycle as 59 days, and the original slide image does show 59 days: this is not an extraction artifact. The reconciliation is rounding. The unrounded periods are DI = 58.717972, DSO = 45.468775 and DPO = 45.558199, giving a cycle of 58.628549 days, which rounds to 59. Adding the periods after rounding them first (59 + 45 − 46) gives 58 instead. Carry the decimals through to the final answer. Treat 58 as correct only when the exercise statement itself supplies the three periods already rounded.

Watch out: the Walmart/Boeing self-test has no official solution in the extracted material (it is left as an in-class exercise) — the figures above are computed by applying the same formulas demonstrated in the solved example, not copied from an answer key.

2. Managing Inventories: The Carrying-Cost vs Order-Cost Trade-off

The trade-off: firms must strike a balance between the benefits and the costs of holding inventory. Benefits: on-time delivery, bulk-purchase discounts, and the ability to meet an unexpected surge in demand. Costs: carrying cost (cost of storage plus the opportunity cost of the capital tied up) and order cost (handling and delivery cost of placing and receiving an order).

The two cost curves move in opposite directions as order size changes: carrying cost rises with order size (larger orders mean more inventory sitting in storage), while order cost falls with order size (fewer, larger orders mean fewer handling/delivery events). The optimal order size sits where the two curves are best balanced, minimizing total cost.

Essential features of inventory management, per the source: optimal inventory levels trade off carrying costs against order costs; a firm can manage inventory by waiting until stock reaches a minimum threshold and then replenishing with a predetermined order quantity (the **reorder point**); when carrying costs are high and order costs are low, it makes sense to place more frequent (smaller) orders; and inventory levels do not rise in direct proportion to sales — as sales grow, the optimal inventory level rises, but less than proportionately (an economy of scale in inventory

management). One inventory strategy worth naming explicitly: **just-in-time (JIT)**, which works only when the flow of production is steady and predictable, so that no significant safety buffer is needed.

> **Watch out:** the source presents the carrying-cost/order-cost trade-off only as a conceptual chart (cost on the y-axis, order size on the x-axis, two intersecting curves) — no explicit algebraic EOQ (economic order quantity) formula is given, so none is reproduced here; know the trade-off logic and the reorder-point mechanism, not a formula.

3. Receivables Management I: Terms of Sale and Credit Analysis

Managing trade credit requires answering four sets of questions: terms of sale, credit analysis, the credit decision, and collection policy. The first two are covered here; the credit decision and collection policy follow in Section 4.

3.1 Terms of Sale

> **Terms of sale:** how long a firm gives customers to pay their bills, and whether it offers a cash discount for prompt payment. Not all sales involve credit at all — some are cash on delivery (COD) or even cash before delivery (CBD).

Each industry has its own logic: firms selling consumer durables may allow a buyer a month to pay, while sellers of perishable goods typically demand much shorter terms. A seller may extend more generous payment terms when its customers operate in a low-risk business, when their accounts are large, when the seller needs time to check the quality of the goods sold, or when the goods are not quickly resold by the buyer.

3.2 Credit Analysis

> **Credit analysis:** the process of determining which customers are likely to pay their bills on time, since not all customers pay by the due date.

For new customers, a firm can check the customer's own financial statements to form its own assessment (financial sustainability ratios, liquidity ratios, etc.), or seek the view of a specialist credit-assessment agency — Dun & Bradstreet or Cerved for smaller companies, Moody's or Standard & Poor's for a firm's bonds. Analytical effort should concentrate on the largest clients in the portfolio, not be spread evenly across all accounts.

4. Receivables Management II: The Credit Decision and Collection Policy

4.1 The Credit Decision

> **The credit decision:** how much credit a firm is prepared to extend to a given customer. If credit is refused, the firm makes neither a profit nor a loss. If credit is granted, there is a probability p that the customer pays (earning revenue − cost) and a probability (1 − p) that the customer defaults (losing the cost).

$$E[\text{gain} \mid \text{refuse credit}] = \$0$$
$$E[\text{gain} \mid \text{grant credit}] = p \times PV(\text{Revenue} - \text{Cost}) - (1-p) \times PV(\text{Cost})$$

where:

- p = the estimated probability the customer pays on time; (1−p) = the estimated probability of default.
- PV = present value; Revenue − Cost is the profit if paid, Cost is the loss if the customer defaults.
- Grant credit only if the expected gain from granting exceeds $0 (the expected gain from refusing).

Three basic principles for exercising judgment in credit allocation, per the source: (1) **maximize profit**, not minimize the number of bad accounts — as credit manager, your job is to maximize present value by weighing the best case (on-time payment) against the worst case (default); a high profit margin justifies a more liberal credit policy, a low margin cannot absorb many bad debts; (2) **concentrate effort on the dangerous accounts** — do not spend equal analytical effort on every credit application, focus on large or doubtful ones; (3) **look beyond the immediate order** — the credit decision is dynamic, and it can be worth accepting a relatively poor risk today if there is a good chance the customer becomes a regular, reliable buyer.

4.2 Collection Policy

> **Collection policy:** the process for collecting money once it becomes due, and the response to reluctant payers or defaulters. Firms typically keep an aging schedule showing what proportion of accounts are mildly or badly overdue.

There can be a useful synergy between the sales and collection departments — for instance, a firm may extend a loan-like arrangement to a customer who has just been cut off by its own bank. Doing so can still make sense even when a bank refuses to lend, provided: (1) the firm has more information than the bank about the customer's business, or (2) the firm needs to look beyond the immediate transaction, recognizing it may lose profitable future sales if the customer goes out of business. Large firms can reap economies of scale in record-keeping, billing and collection; a small firm that cannot support a full credit operation may instead benefit from farming out part of the job to a factor.

5. Common Mistakes & Exam Traps

- Mixing up the denominators: DSO is scaled by daily **sales**; DI and DPO are both scaled by daily **COGS** — using sales for all three is a common error.
- Forgetting the sign in the cash cycle formula: Cash Cycle = DI + DSO − DPO (subtract the payable period, don't add it).
- Assuming inventory should scale proportionally with sales — the source is explicit that optimal inventory rises less than proportionately as sales grow.
- Treating the credit decision as "avoid all bad debts" rather than "maximize expected present value" — the source explicitly warns against minimizing the number of bad accounts as an objective in itself.
- Believing that shortening customer payment terms is a costless way to plug a supplier-driven liquidity gap: the exam scenario built directly on this (a company facing a €20 million liquidity need from a supplier's unilateral term change, responding by shortening customer terms by 10 days) has a value-destruction outcome as its answer — lost customers and revenue outweigh the

cash captured, echoing Mistake 4 from Session 2.

- Assuming that improving the current ratio and quick ratio is unambiguously good treasury practice — doing so typically means accumulating more receivables and inventory, which reduces capital efficiency rather than improving it (same logic as Mistake 5 in Session 2).
- Assuming a strong quality reputation justifies a longer production cycle without checking whether the market actually pays for it — excess focus on defect-reduction quality metrics ties up capital in WIP inventory without a proportional return if customers won't pay a premium for it.

6. Must Know Absolutely

Must know:
• DSO = AR ÷ (Sales ÷ 365); DPO = AP ÷ (COGS ÷ 365); DI = Inventory ÷ (COGS ÷ 365).
• Cash Cycle = DI + DSO − DPO = operating cycle (DI+DSO) minus the payable period.
• Inventory optimum trades carrying cost (storage + opportunity cost of capital) against order cost (handling/delivery); replenish at the reorder point.
• Optimal inventory rises less than proportionately with sales; JIT only works with steady, predictable production flow.
• Four receivables questions: terms of sale, credit analysis, the credit decision, collection policy.
• Credit decision: E[gain|grant] = p×PV(Rev−Cost) − (1−p)×PV(Cost); goal is to maximize PV, not minimize bad debts.
• Concentrate credit-analysis effort on large/doubtful accounts.
• Aging schedule tracks overdue receivables; extending credit past a bank refusal can be rational with an information edge or to protect future sales.

Watch Out

Watch out: the solved cash-cycle example's final answer of 59 days is genuine (verified on the original slide image): computed on the unrounded periods it is 58.628549 ≈ 59. The 58 that comes out of 59 + 45 − 46 is an artifact of rounding each period before summing, not of the formula and not of the slide. An earlier version of this fiche misread the 59 as a PDF-extraction artifact. It is not one.

Watch out: the Walmart/Boeing self-test carries no official answer key in the extracted slides; the values given here follow directly from the formulas demonstrated in the solved example, applied to the source's own numbers.

Watch out: the inventory cost-trade-off chart (order cost vs carrying cost vs order size) is a visual diagram in the original slide with no accompanying algebraic formula — only the qualitative trade-off is testable from this material.